Regimes

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1. Motivation

My historical research looks at the role of <u>conditioning</u> information in portfolio selection

- Early work by Markowitz assumed exact knowledge static expected returns (just use long-term averages)
- My work generalized his framework to include both time-varying returns, volatilities, correlations and skew
- Getting the model right is fraught with the danger of overfitting

The Journal of FINANCE

Conditional Skewness in Asset Pricing Tests

CAMPBELL R. HARVEY and AKHTAR SIDDIQUE*

ABSTRACT

If asset returns have systematic skewness, expected returns should include rewards for accepting this risk. We formalize this intuition with an asset pricing

Plan

1. Motivation
2. Four state MTP model
3. Multi-state model
   a) Factor application
   b) FX application
4. Portfolio formation: Base Pairs
5. Questions

2. Four State Regime Model

Momentum Turning Points

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Momentum turning points

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ABSTRACT

We use slow and fast time-series momentum to characterize four stock market cycles—Bull, Correction, Bear, and Rebound. The steep market declines of Bears concentrate in high-risk states, yet predict negative expected returns, which is difficult to rationalize by most models of time-varying risk premia. Using a model to analyze slow and fast momentum strategies, we estimate both relatively high mean persistence and realization noise in U.S. stock market returns. Intermediate-speed momentum portfolios, formed by blending slow and fast momentum strategies, translate predictive information in market cycles into positive unconditional alpha, for which we propose a novel decomposition.

Breaking Bad Trends

Financial Analysts Journal | A Publication of CFA Institute
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Research

Breaking Bad Trends

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Setting

Two types of mistakes:

- If a trend-following system is **too slow**, you risk a Type II error by missing a turning point
- If a trend-following system is **too fast**, you risk a Type I error by reacting to noise
- The Holy Grail of trend following is a dynamic system to adjust speed depending on market/economic conditions.

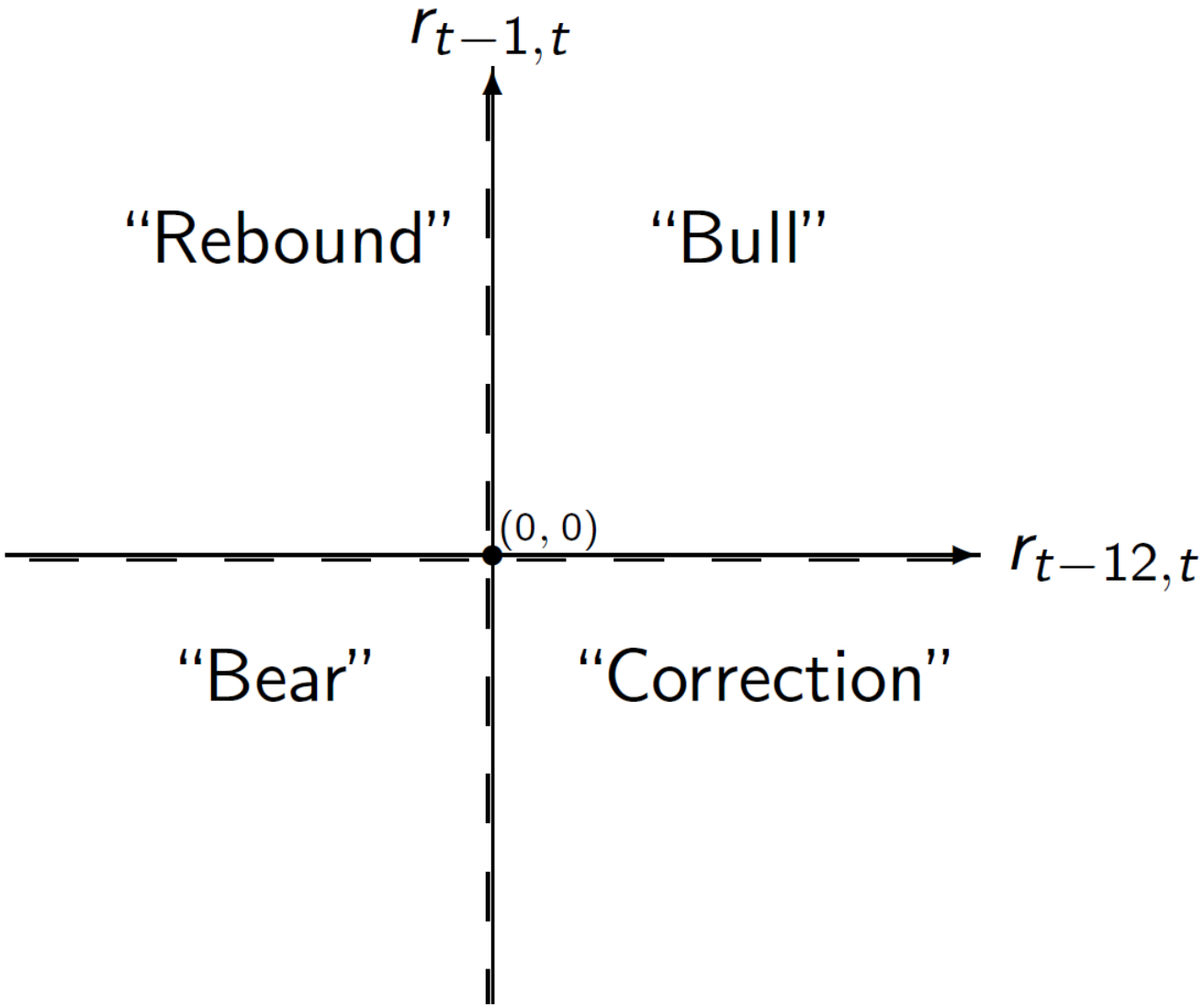
Setting

Contributions:

- Document the impact of breaks on the performance of trend-following strategy
- Detail a way to develop conditionally expected returns that depend on four states (not two – up phase and down phase)
- Dynamic weighting of fast and slow speeds

Four States

- Four observable market regimes:

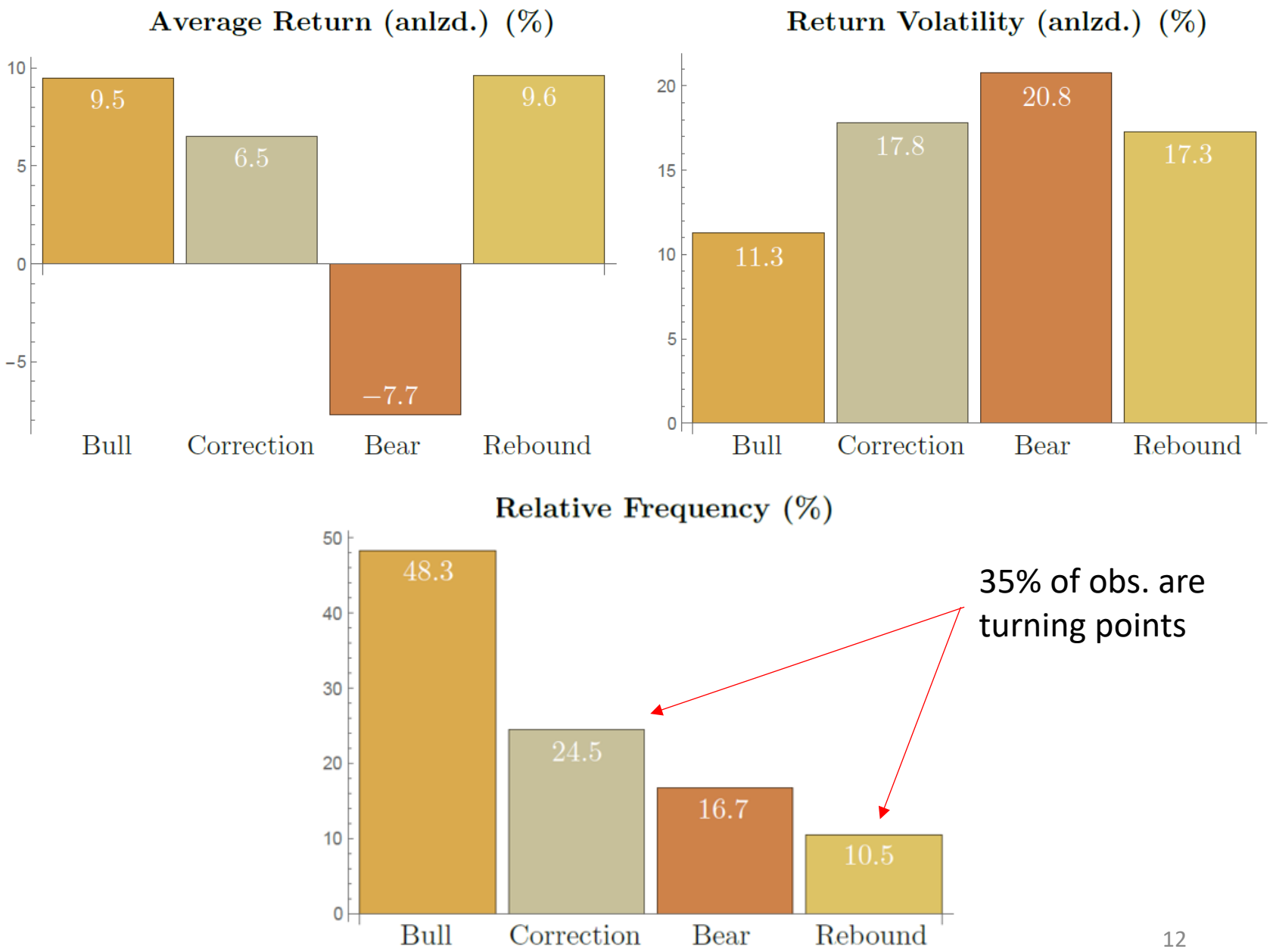


- **All strategies generated ex ante**

S&P 500

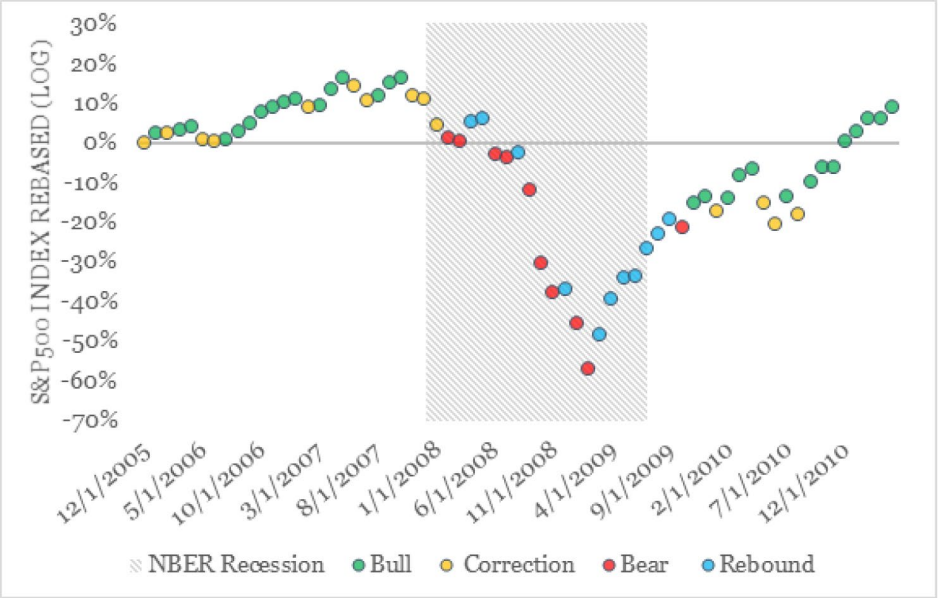
- All summary statistics generated ex ante
- US data over 50 years

Figure 1: **U.S. Stock Market Cycles**



MTP Conclusions

- MTP successfully applied to many different markets and factors
- Static MTP requires no parameter estimates – the world is divided into four states
- Dynamic MTP requires some burn in
  - Slow down after corrections and speed up after rebounds



- <u>What happens if we expand the state space</u>?

3a): Expanding the State Space

New research in progress with **Otto van Hemert** and **Amara Mulliner**

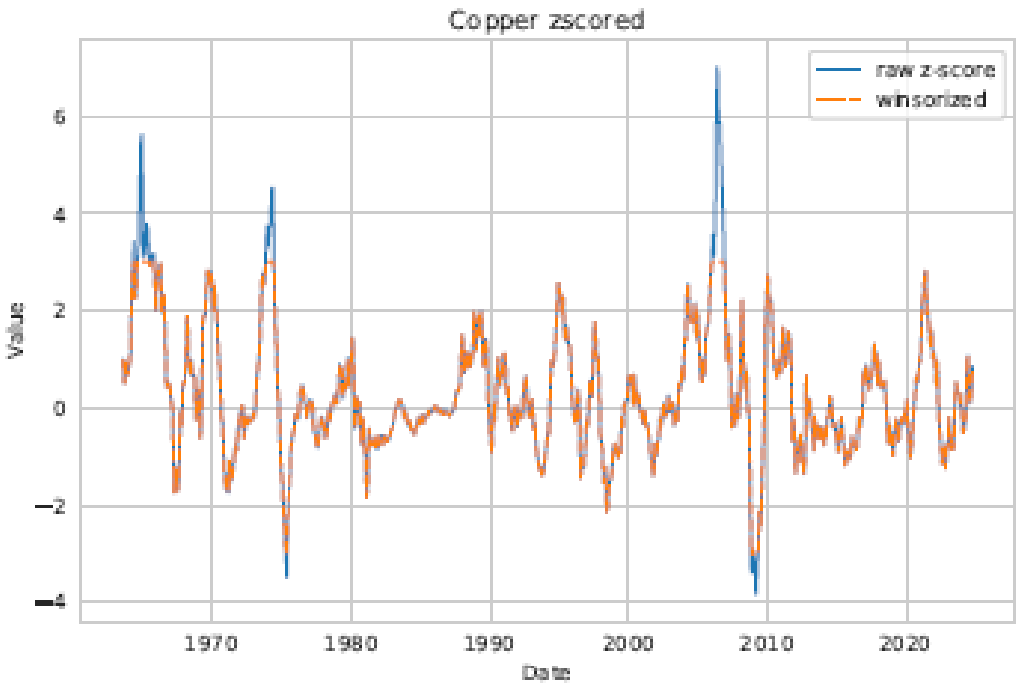
- Expand the set of economic variables

Variable	Proxy	Data source	Start date
Market	S&P500	Bloomberg	1927
Yield curve	US 10-year yield minus US 3-month yield	Internal data & St Louis FRED	1962
Oil	WTI Spot Crude Oil Price	St Louis FRED	1946
Copper	Copper futures price	Internal data	1959
Monetary policy	US 3-month yield	St Louis FRED	1954
Volatility	Realised volatility of S&P500 stitched with the VIX Index (from 1990)	Bloomberg	1929
Stock-bond correlation	S&P500 and US 10 year yield	Bloomberg & internal data	1962

Expanding the State Space

Transform all variables (some have trends)

- Z-score and winsorize at 3 sigma



Expanding the State Space

Similarity

- At each date and for each variable, we measure <u>how similar</u> the variable is to each historical month (use squared distance).
- So, if z-score for a variable in February 2025 = 2.5. We look for all historical dates with a score very close to 2.5.
- We do this for all seven variables and sum across variables (then take square root)
- So looking at the state today, we can look historically to see the <u>dates that are the most similar to today</u>.
- We then look at the <u>historical return AFTER the similar dates</u>
- This informs our trading position

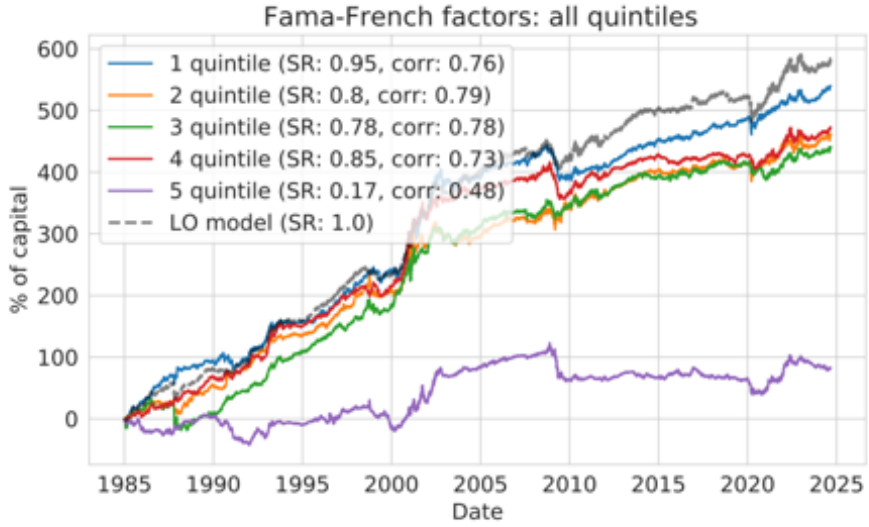
Expanding the State Space

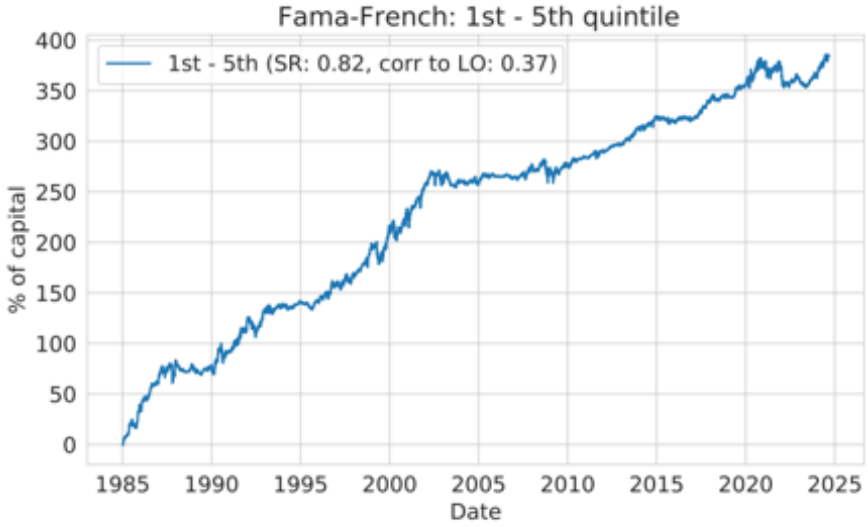
Preliminary results

- Like MTP, the method is highly non-parametric

Exhibit 9: Assessing the predictability using six long-short stock factors

The performance of investing in six long-short stock factors (Market, Size, Value, Profitability, Investment, Momentum). For the quintile 1 portfolio, the direction taking in a factor is based on returns subsequent to the 20% most similar historical dates. The quintile 5 portfolio utilizes the 20% most dissimilar returns. The long-only portfolio is long all six factors throughout. Performance is shown for 1985-2024. Input data starts well before 1985 to allow for rolling-window calibrations.





3b): Regimes in FX

New research with:

Otto van Hemert, Pedro Castro, Carl Hamill, John Harber

- Often an asset owner with international equity holdings faces the question: <u>To hedge or not to hedge?</u>
- We argue that hedging should not be thought of as a static choice: <u>conditioning information</u> should play a role
- One obvious piece of conditioning information is the interest rate differential between two countries or <u>carry</u>:
  - Japanese investors (low interest rates) on average better off not hedging
  - NZ investors (high interest rates) on average better off hedging

3b): Regimes in FX

The Best Strategies for FX Hedging

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Abstract

The question of whether, when, and how to hedge foreign exchange risk has been a vexing one for investors since the end of the Bretton Woods system in 1973. Our study provides a comprehensive empirical analysis of dynamic FX hedging strategies over several decades, examining various domestic and foreign currency pairs. While traditional approaches often focus on risk mitigation, we explore the broader implications for expected returns, highlighting the interplay between hedging and strategies such as the carry trade. Our findings reveal that incorporating additional factors-such as trend (12-month FX return), value (deviation from purchasing power parity), and carry (interest rate differential) - into hedging decisions delivers significant portfolio benefits. By adopting a dynamic, active approach to FX hedging, investors can enhance returns and manage risk more effectively than with static hedged or unhedged strategies.

https://papers.ssrn.com/sol3/papers.cfm?al

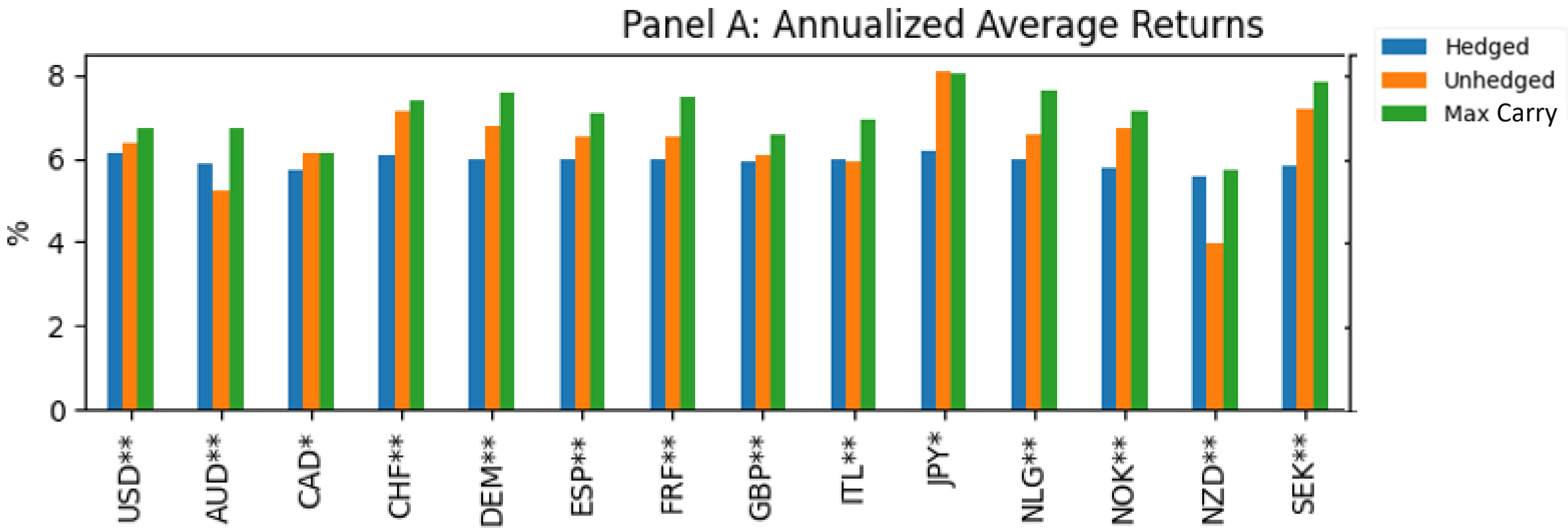
Regimes in FX

	% of Months with Positive IRD vs USD	Average FX Return when IRD Positive	Average FX Return when IRD Negative	Average World Equity Return (Hedged)	Average World Equity Return (Unhedged)	Average World Equity Return (Max Carry)
DM						
USD	--	--	--	5.4%	5.6%	6.0%
AUD	77.4%	4.0%	-5.9%	5.8%	5.3%	6.8%
CAD	75.6%	0.4%	-0.6%	5.0%	5.4%	5.5%
CHF	13.9%	1.6%	0.6%	5.3%	5.9%	6.1%
DEM	30.3%	4.8%	-1.2%	5.2%	5.7%	6.5%
ESP	63.3%	2.7%	-2.8%	5.2%	5.6%	6.3%
FRF	59.1%	4.4%	-4.1%	5.2%	5.3%	6.6%
GBP	77.0%	1.9%	-3.2%	5.2%	5.4%	5.8%
ITL	66.7%	2.6%	-3.2%	5.2%	5.5%	6.2%
JPY	21.8%	1.8%	-1.4%	5.4%	7.2%	7.2%
NLG	33.3%	5.8%	-1.8%	5.2%	5.5%	6.7%
NOK	59.9%	2.6%	-3.0%	5.0%	5.9%	6.3%
NZD	89.3%	2.8%	0.7%	4.8%	3.8%	5.0%
SEK	59.4%	2.9%	-4.5%	5.1%	6.4%	7.0%

Low rate home currency: no hedging

High rate home currency: hedging

Regimes in FX



Regimes in FX

Exercise does not take portfolio considerations into account

- Min vol: Uses all equities and FX, to form min vol hedge ratios (ratios restricted to range from [0,1]
- Optimal: Uses risk aversion of 4/3, to incorporate carry
- Mom: Uses information in momentum
- PPP: Uses “value” signal
- Blend: Average of active strategies

Regimes in FX

Average Returns:

	Unhedged	Hedged	Min Vol	Max Carry	Optimal	Mom	PPP	Blend
USD	6.4%	6.1%	6.3%	6.7%	6.6%	6.9%	6.4%	6.6%
AUD	5.2%	5.9%	4.9%	6.7%	6.1%	5.3%	6.8%	6.0%
CAD	6.1%	5.7%	5.9%	6.1%	6.0%	6.5%	6.6%	6.4%
CHF	7.2%	6.1%	6.2%	7.4%	7.7%	7.1%	7.4%	7.4%
DEM	6.8%	6.0%	6.4%	7.6%	7.3%	6.9%	6.9%	7.1%
ESP	6.5%	6.0%	6.8%	7.1%	6.7%	7.0%	6.5%	6.7%
FRF	6.5%	6.0%	6.6%	7.5%	7.3%	6.8%	6.7%	6.9%
GBP	6.1%	5.9%	6.5%	6.6%	6.7%	6.8%	6.9%	6.8%
ITL	5.9%	6.0%	6.6%	6.9%	6.7%	6.1%	6.4%	6.4%
JPY	8.1%	6.2%	6.4%	8.0%	8.1%	8.8%	7.5%	8.1%
NLG	6.6%	6.0%	6.4%	7.6%	7.5%	6.7%	7.0%	7.1%
NOK	6.7%	5.8%	6.3%	7.1%	7.0%	6.7%	7.4%	7.0%
NZD	4.0%	5.6%	4.7%	5.7%	5.6%	5.8%	6.1%	5.8%
SEK	7.2%	5.8%	6.6%	7.9%	7.2%	7.1%	7.5%	7.3%

Regimes in FX

Sharpe ratios:

	Unhedged	Hedged	Min Vol	Max Carry	Optimal	Mom	PPP	Blend
USD	0.44	0.45	0.46	0.48	0.48	0.49	0.46	0.48
AUD	0.37	0.42	0.38	0.50	0.46	0.38	0.50	0.46
CAD	0.48	0.42	0.47	0.46	0.47	0.50	0.50	0.50
CHF	0.44	0.45	0.46	0.47	0.51	0.47	0.47	0.49
DEM	0.44	0.44	0.48	0.51	0.52	0.48	0.49	0.50
ESP	0.43	0.44	0.51	0.51	0.50	0.48	0.48	0.50
FRF	0.43	0.44	0.50	0.53	0.53	0.47	0.47	0.50
GBP	0.41	0.43	0.49	0.48	0.50	0.49	0.49	0.50
ITL	0.40	0.44	0.50	0.50	0.50	0.42	0.47	0.47
JPY	0.49	0.46	0.47	0.50	0.54	0.58	0.48	0.54
NLG	0.43	0.44	0.48	0.52	0.54	0.46	0.49	0.51
NOK	0.47	0.42	0.49	0.51	0.53	0.48	0.51	0.52
NZD	0.27	0.40	0.36	0.42	0.43	0.42	0.45	0.45
SEK	0.49	0.42	0.50	0.57	0.55	0.51	0.52	0.54

Regimes in FX

Hedge or unhedged performs worst in every country

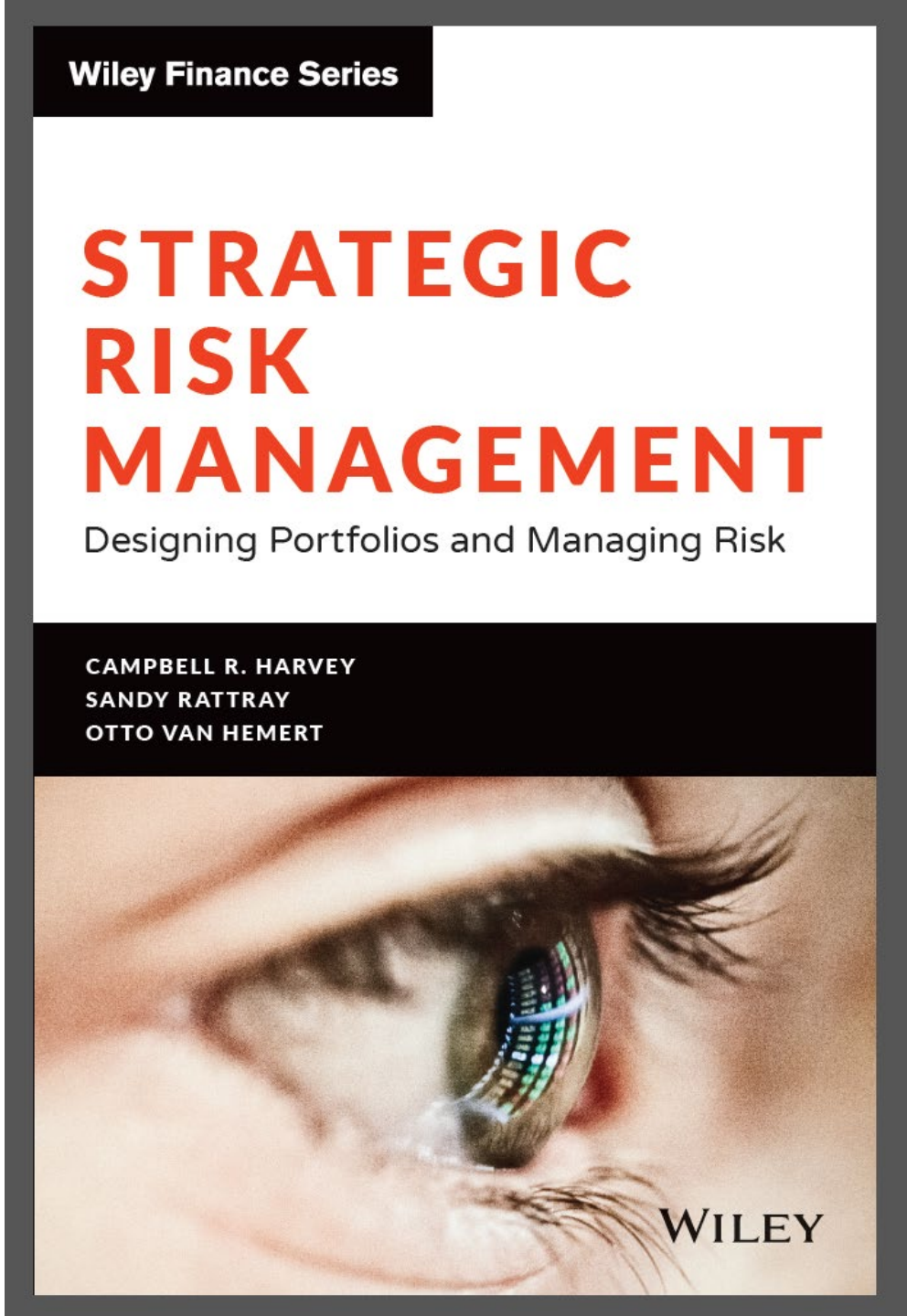
Sharpe ratios:

	Unhedged	Hedged	Min Vol	Max Carry	Optimal	Mom	PPP	Blend
USD	0.44	0.45	0.46	0.48	0.48	0.49	0.46	0.48
AUD	0.37	0.42	0.38	0.50	0.46	0.38	0.50	0.46
CAD	0.48	0.42	0.47	0.46	0.47	0.50	0.50	0.50
CHF	0.44	0.45	0.46	0.47	0.51	0.47	0.47	0.49
DEM	0.44	0.44	0.48	0.51	0.52	0.48	0.49	0.50
ESP	0.43	0.44	0.51	0.51	0.50	0.48	0.48	0.50
FRF	0.43	0.44	0.50	0.53	0.53	0.47	0.47	0.50
GBP	0.41	0.43	0.49	0.48	0.50	0.49	0.49	0.50
ITL	0.40	0.44	0.50	0.50	0.50	0.42	0.47	0.47
JPY	0.49	0.46	0.47	0.50	0.54	0.58	0.48	0.54
NLG	0.43	0.44	0.48	0.52	0.54	0.46	0.49	0.51
NOK	0.47	0.42	0.49	0.51	0.53	0.48	0.51	0.52
NZD	0.27	0.40	0.36	0.42	0.43	0.42	0.45	0.45
SEK	0.49	0.42	0.50	0.57	0.55	0.51	0.52	0.54

4) Regimes in Portfolio Formation: Base pairs

New research "Investment Base Pairs" with **Christian Goulding**

- Long literature on cross-sectional strategies
- Asset managers tend to use traditional approaches such as sorting or transforming signal into weight
- Sometimes performance deteriorates: is this due to signal failing – or a result of how the portfolio is constructed?
- We take a fresh look and propose a new approach: Base Pairs

Papers available

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